

Fatma Ayad

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EDUCATION

Bryn Mawr College

Bachelor of Arts in Computer Science, Minor in Data Science — GPA: 3.830

Bryn Mawr, PA

Expected Graduation: Spring 2028

- Activities: President of Girls Who Code, Break Through Tech Fellow, FIRST Robotics, Grace Hopper, Codepath
- Relevant Coursework: Algorithms, Data Science, Systems Programming, Cloud Computing, Databases

University of Pennsylvania

Incoming Master of Science in Engineering in Computer and Information Science

Philadelphia, PA

Expected Graduation: Spring 2029

TECHNICAL SKILLS

Languages: Java, Python, C, C++, HTML/CSS, JavaScript, TypeScript, Swift, SQL

Libraries/Frameworks: React, Node.js, Next.js, D3.js, Pandas, NumPy, AstroPy, Matplotlib, Pygame, NCurses, JUnit

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Frizzle

New York, NY

- Built a **LangSmith**-based evaluation pipeline processing **15K+** traces across 8 production AI workflows, establishing an automated grading framework extensible to new subjects; shipped a debugging tool to production that surfaces root-cause errors in failed traces, cutting error diagnosis time by **40%**.
- Engineered a 4-stage **RAG** system to extract student concept mastery from student work, scraping and embedding **2K+** concepts across math, English, and social sciences into a **Supabase** vector database; achieved **95%** precision/recall on semantic concept matching, improving mastery detection accuracy by **40%** over keyword-based baselines.

Machine Learning Research Assistant

June 2025 – Present

Bryn Mawr College

Bryn Mawr, PA

- Developed a large-scale **LLM benchmark** by transforming 1200+ GitHub PR comments into 30 pair-programming questions across Java, C++, and Python enabling cross-model benchmarking and revealing gaps in contextualized code-reasoning.
- Designed a numeric scoring rubric standardized across **20+ LLMs**, then automated rubric applications with **Claude Opus 4** validated against human scoring to reach **94% agreement**, cutting assessment time per response by **70%**.
- Deployed full-stack benchmark visualization platform using React, Next.js and D3.js, enabling 100+ researchers and FAANG engineers in Google and Meta to explore our benchmark's results across 6+ metrics and 20+ model variants.

Software Engineering Intern

May 2026 – Present

Develop for Good - Ikhaya Lethu Care for the Elderly

Remote

- Engineered a full-stack, responsive website (**React**, **Next.js**, **TypeScript**, **Tailwind CSS**) for a non-profit, with role-based access across **admin**, **vendor**, and **donor** roles, reaching **200–300** seniors and **50–100** vendors in South Africa.
- Built a **Node.js** backend and RESTful API enabling vendors to self-register and publish/edit product listing pages, reducing manual onboarding time by **30%** and cutting vendor listing setup time from **50+** to **15** minutes.

Digital Technology Intern

May – August 2025

Bryn Mawr College

Bryn Mawr, PA

- Led a **5-member** team to convert **5 years** of institutional data from static PDF tables into **20+** interactive Tableau dashboards, optimizing performance and cross-browser rendering while embedding them into the College's website via **HTML**, increasing accessibility for **400+** regular users across **10** institutional categories.

PUBLICATIONS

Ferida Mohammed, **Fatma Ayad**, Satish Chandra, Petros Maniatis, Elizabeth Dinella.

"RubberDuckBench: A Benchmark for AI Coding Assistants."

Published at the International Conference on Software Engineering (ICSE), accepted, 2025.

TECHNICAL PROJECTS

LiveBench Refresh | Python, Claude API, Embeddings, LLM-as-Judge

- Built a benchmark refresh pipeline detecting data contamination in LLM benchmark questions via **membership inference**, flagging **25%** of **30+** questions across [**SWE-Bench**, **RubberDuckBench**] as compromised.
- Designed a semantic-preserving LLM-based paraphrasing technique to restructure compromised questions, reducing similarity to originals by **10-15%** while preserving **95%** rubric-scored difficulty equivalence.

Room8 | Swift, SwiftUI, MVVM Architecture

Google's Gemini Hackathon

- Developed a full-stack iOS roommate management app using **MVVM** architecture to track shared expenses, chores, and household responsibilities across **10+** users, replacing scattered group chats with a unified system.
- Integrated **Gemini Vision API** to auto-parse receipt photos into itemized shared expenses with **85%** accuracy, eliminating manual entry and cutting expense-logging time by **20%**.